

TIC 399770041

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.580189 d
Epoch (t0) 1569.9907 BTJD
Duration 0.0840 d
Depth 4.291 % (42,906 ppm)
SNR 488.6

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 30 (gate ≥ 3)
Robust depth ratio 1.04 (gate ≥ 0.50)
Robust depth sigma 283.6 (gate ≥ 7.0)
Odd/even z-score 0.86 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 10 light-curve dip in this sector: 202.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 36.92 (gate ≥ 5.0) Compactness 0.88 (gate ≥ 0.50)
Dimming centroid offset 0.04 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 399770066
1.83 px away, dTmag +2.10 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

